

Chapter 41: Pancreatic Hormones & Antidiabetic Drugs**MEDICATIONS FOR TREATMENT OF TYPE 2 DIABETES**

Several categories of glucose-lowering agents are available for patients with type 2 diabetes: (1) agents that bind to the sulfonylurea receptor and stimulate insulin secretion (sulfonylureas, meglitinides, D-phenylalanine derivatives); (2) agents that lower glucose levels by their actions on liver, muscle, and adipose tissue (biguanides, thiazolidinediones); (3) agents that principally slow the intestinal absorption of glucose (α -glucosidase inhibitors); (4) agents that mimic incretin effect or prolong incretin action (GLP-1 receptor agonists, dipeptidyl peptidase 4 [DPP-4] inhibitors), (5) agents that inhibit the reabsorption of glucose in the kidney (sodium-glucose co-transporter inhibitors [SGLTs]), and (6) agents that act by other or ill-defined mechanisms (pramlintide, bromocriptine, colesevelam).

DRUGS THAT PRIMARILY STIMULATE INSULIN RELEASE BY BINDING TO THE SULFONYLUREA RECEPTOR**SULFONYLUREAS****Mechanism of Action**

The major action of sulfonylureas is to increase insulin release from the pancreas (Table 41-7). They bind to a 140-kDa high-affinity sulfonylurea receptor that is associated with a beta-cell inward rectifier ATP-sensitive potassium channel (Figure 41-2). Binding of a sulfonylurea inhibits the efflux of potassium ions through the channel and results in depolarization. Depolarization opens a voltage-gated calcium channel and results in calcium influx and the release of preformed insulin.

TABLE 41-7

Regulation of insulin release in humans.

Stimulants of insulin release
Humoral: Glucose, mannose, leucine, arginine, other amino acids, fatty acids (high concentrations)
Hormonal: Glucagon, glucagon-like peptide 1 (7-37), glucose-dependent insulinotropic polypeptide, cholecystokinin, gastrin
Neural: β -Adrenergic stimulation, vagal stimulation
Drugs: Sulfonylureas, meglitinide, nateglinide, isoproterenol, acetylcholine
Inhibitors of insulin release
Hormonal: Somatostatin, insulin, leptin
Neural: α -Sympathomimetic effect of catecholamines
Drugs: Diazoxide, phenytoin, vinblastine, colchicine

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Efficacy & Safety of the Sulfonylureas

Sulfonylureas are metabolized by the liver and, with the exception of acetohexamide, the metabolites are either weakly active or inactive. The metabolites are excreted by the kidney and, in the case of the second-generation sulfonylureas, partly excreted in the bile. Idiosyncratic reactions are rare, with skin rashes or hematologic toxicity (leukopenia, thrombocytopenia) occurring in less than 0.1% of cases. The second-generation sulfonylureas have greater affinity for their receptor compared with the first-generation agents. The correspondingly lower effective doses and plasma levels of the second-generation drugs therefore lower the risk of drug-drug interactions based on competition for plasma binding sites or hepatic enzyme action.

In 1970, the University Group Diabetes Program (UGDP) in the United States reported that the number of deaths due to cardiovascular disease in diabetic patients treated with tolbutamide was excessive compared with either insulin-treated patients or those receiving placebos. Owing to design flaws, this study and its conclusions were not generally accepted. In the United Kingdom, the UKPDS did not find an untoward cardiovascular effect of sulfonylurea usage in their large, long-term study. The sulfonylureas continue to be widely prescribed, and six are available in the United States.

FIRST-GENERATION SULFONYLUREAS

Tolbutamide is well absorbed but rapidly metabolized in the liver. Its duration of effect is relatively short (6-10 hours), with an elimination half-life of 4-5 hours, and it is best administered in divided doses (eg, 500 mg before each meal). Some patients only need one or two tablets daily. The maximum dosage is 3000 mg daily. Because of its short half-life and inactivation by the liver, it is relatively safe in the elderly and in patients with renal impairment. Prolonged hypoglycemia has been reported rarely, mostly in patients receiving certain antibacterial sulfonamides (sulfisoxazole), phenylbutazone for arthralgias, or the oral azole antifungal medications to treat candidiasis. These drugs inhibit the metabolism of tolbutamide in the liver and increase its circulating levels.

Chlorpropamide has a half-life of 32 hours and is slowly metabolized in the liver to products that retain some biologic activity; approximately 20-30% is excreted unchanged in the urine. The average maintenance dosage is 250 mg daily, given as a single dose in the morning. Prolonged hypoglycemic reactions are more common in elderly patients, and the drug is contraindicated in this group. Other adverse effects include a hyperemic flush after

alcohol ingestion in genetically predisposed patients and hyponatremia due to its effect on vasopressin secretion and action.

Tolazamide is comparable to chlorpropamide in potency but has a shorter duration of action. Tolazamide is more slowly absorbed than the other sulfonylureas, and its effect on blood glucose does not appear for several hours. Its half-life is about 7 hours. Tolazamide is metabolized to several compounds that retain hypoglycemic effects. If more than 500 mg/d are required, the dosage should be divided and given twice daily.

Acetohexamide is no longer available in the United States. Its half-life is only about 1 hour but its more active metabolite, hydroxyhexamide, has a half-life of 4–6 hours; thus the drug duration of action is 8–24 hours. Where available, its dosage is 0.25–1.5 g/d as single dose or in two divided doses.

Chlorpropamide, tolazamide, and acetohexamide are now rarely used in clinical practice.

SECOND-GENERATION SULFONYLUREAS

Glyburide, glipizide, gliclazide, and glimepiride are 100–200 times more potent than tolbutamide. They should be used with caution in patients with cardiovascular disease or in elderly patients, in whom hypoglycemia would be especially dangerous.

Glyburide is metabolized in the liver into products with very low hypoglycemic activity. The usual starting dosage is 2.5 mg/d or less, and the average maintenance dosage is 5–10 mg/d given as a single morning dose; maintenance dosages higher than 20 mg/d are not recommended. A formulation of “micronized” glyburide (Glynase PresTab) is available in a variety of tablet sizes. However, there is some question as to its bioequivalence with non-micronized formulations, and the FDA recommends careful monitoring to re-titrate dosage when switching from standard glyburide doses or from other sulfonylurea drugs.

Glyburide has few adverse effects other than its potential for causing hypoglycemia. Flushing has rarely been reported after ethanol ingestion, and the compound slightly enhances free water clearance. Glyburide is contraindicated in the presence of hepatic impairment and in patients with renal insufficiency.

Glipizide has the shortest half-life (2–4 hours) of the more potent agents. For maximum effect in reducing postprandial hyperglycemia, this agent should be ingested 30 minutes before breakfast because absorption is delayed when the drug is taken with food. The recommended starting dosage is 5 mg/d, with up to 15 mg/d given as a single dose. When higher daily dosages are required, they should be divided and given before meals. The maximum total daily dosage recommended by the manufacturer is 40 mg/d, although some studies indicate that the maximum therapeutic effect is achieved by 15–20 mg of the drug. An extended-release preparation (Glucotrol XL) provides 24-hour action after a once-daily morning dose (maximum of 20 mg/d). However, this formulation appears to have sacrificed its lower propensity for severe hypoglycemia compared with longer-acting glyburide without showing any demonstrable therapeutic advantages over the latter (which can be obtained as a generic drug). At least 90% of glipizide is metabolized in the liver to inactive products, and the remainder is excreted unchanged in the urine. Glipizide therapy is therefore contraindicated in patients with significant hepatic impairment. Because of its lower potency and shorter duration for action, it is preferable to glyburide in the elderly and for those patients with renal impairment.

Glimepiride is approved for once-daily use as monotherapy or in combination with insulin. Glimepiride achieves blood glucose lowering with the lowest dosage of any sulfonylurea compound. A single daily dose of 1 mg has been shown to be effective, and the recommended maximal daily dosage is 8 mg. Glimepiride’s half-life under multidose conditions is 5–9 hours. It is completely metabolized by the liver to metabolites with weak or no activity.

Gliclazide (not available in the United States) has a half-life of 10 hours. The recommended starting dosage is 40–80 mg daily with a maximum dosage of 320 mg daily. Higher dosages are usually divided and given twice a day. It is completely metabolized by the liver to inactive metabolites.

MEGLITINIDE ANALOGS

Repaglinide is the first member of the meglitinide group of insulin secretagogues. These drugs modulate beta-cell insulin release by regulating potassium efflux through the potassium channels previously discussed. There is overlap with the sulfonylureas in their molecular sites of action because the meglitinides have two binding sites in common with the sulfonylureas and one unique binding site.

Repaglinide has a fast onset of action, with a peak concentration and peak effect within approximately 1 hour after ingestion, but the duration of action is 4–7 hours. It is cleared by hepatic CYP3A4 with a plasma half-life of 1 hour. Because of its rapid onset, repaglinide is indicated for use in controlling postprandial glucose excursions. The drug should be taken just before each meal in doses of 0.25–4 mg (maximum 16 mg/d); hypoglycemia is a risk if

the meal is delayed or skipped or contains inadequate carbohydrate. It can be used in patients with renal impairment and in the elderly. Repaglinide is approved as monotherapy or in combination with biguanides. There is no sulfur in its structure, so repaglinide may be used in type 2 diabetics with sulfur or sulfonyleurea allergy.

Mitiglinide (not available in the United States) is a benzylsuccinic acid derivative that binds to the sulfonyleurea receptor and is similar to repaglinide in its clinical effects. It has been approved for use in Japan.

D-PHENYLALANINE DERIVATIVE

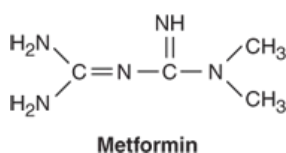
Nateglinide, a D-phenylalanine derivative, stimulates rapid and transient release of insulin from beta cells through closure of the ATP-sensitive K^+ channel. It is absorbed within 20 minutes after oral administration with a time to peak concentration of less than 1 hour and is metabolized in the liver by CYP2C9 and CYP3A4 with a half-life of about 1 hour. The overall duration of action is about 4 hours. It is taken before the meal and reduces the postprandial rise in blood glucose levels. It is available as 60- and 120-mg tablets. The lower dose is used in patients with mild elevations in HbA_{1c} .

Nateglinide is efficacious when given alone or in combination with non-secretagogue oral agents (such as metformin). Hypoglycemia is the main adverse effect. It can be used in patients with renal impairment and in the elderly.

DRUGS THAT PRIMARILY LOWER GLUCOSE LEVELS BY THEIR ACTIONS ON THE LIVER, MUSCLE, & ADIPOSE TISSUE

BIGUANIDES

The structure of **metformin** is shown below. Phenformin (an older biguanide) was discontinued in the United States because of its association with lactic acidosis. Metformin is the only biguanide currently available in the United States.



Mechanisms of Action

A full explanation of the mechanism of action of the biguanides remains elusive, but their primary effect is to activate the enzyme AMP-activated protein kinase (AMPK) and reduce hepatic glucose production. Patients with type 2 diabetes have considerably less fasting hyperglycemia as well as lower postprandial hyperglycemia after administration of biguanides; however, hypoglycemia during biguanide therapy is rare. These agents are therefore more appropriately termed “euglycemic” agents.

Metabolism & Excretion

Metformin has a half-life of 1.5–3 hours, is not bound to plasma proteins, is not metabolized, and is excreted by the kidneys as the active compound. As a consequence of metformin's blockade of gluconeogenesis, the drug may impair the hepatic metabolism of lactic acid. In patients with renal insufficiency, the biguanide accumulates and thereby increases the risk of lactic acidosis, which appears to be a dose-related complication. Metformin can be safely used in patients with estimated glomerular filtration rates (eGFR) between 60 and 45 mL/min per 1.73 m². It can be used cautiously in patients with eGFR between 45 and 30 mL/min per 1.73 m². It is contraindicated if the eGFR is less than 30 mL/min per 1.73 m².

Clinical Use

Biguanides are recommended as first-line therapy for type 2 diabetes. Because metformin is an insulin-sparing agent and does not increase body weight or provoke hypoglycemia, it offers obvious advantages over insulin or sulfonyleureas in treating hyperglycemia in such persons. The UKPDS reported that metformin therapy decreases the risk of macrovascular as well as microvascular disease; this is in contrast to the other therapies, which only modified microvascular morbidity. Biguanides are also indicated for use in combination with insulin secretagogues or thiazolidinediones in type 2 diabetics in whom oral monotherapy is inadequate. Metformin is useful in the prevention of type 2 diabetes; the landmark Diabetes Prevention Program concluded that metformin is efficacious in preventing the new onset of type 2 diabetes in middle-aged, obese persons with impaired glucose

tolerance and fasting hyperglycemia. It is interesting that metformin did not prevent diabetes in older, leaner prediabetics.

Although the recommended maximal dosage is 2.55 g daily, little benefit is seen above a total dosage of 2000 mg daily. Treatment is initiated at 500 mg with a meal and increased gradually in divided doses. Common schedules would be 500 mg once or twice daily increased to 1000 mg twice daily. The maximal dosage is 850 mg three times a day. Epidemiologic studies suggest that metformin use may reduce the risk of some cancers. These data are still preliminary, and the speculative mechanism of action is a decrease in insulin (which also functions as a growth factor) levels as well as direct cellular effects mediated by AMPK. Other studies suggest a reduction in cardiovascular deaths in humans and an increase in longevity in mice (see Chapter 60).

Toxicities

The most common toxic effects of metformin are gastrointestinal (anorexia, nausea, vomiting, abdominal discomfort, and diarrhea), occurring in up to 20% of patients. They are dose related, tend to occur at the onset of therapy, and are often transient. However, metformin may have to be discontinued in 3–5% of patients because of persistent diarrhea.

Metformin interferes with the calcium-dependent absorption of vitamin B₁₂–intrinsic factor complex in the terminal ileum, and vitamin B₁₂ deficiency can occur after many years of metformin use. Periodic screening for vitamin B₁₂ deficiency should be considered, especially in patients with peripheral neuropathy or macrocytic anemia. Increased intake of calcium may prevent the metformin-induced B₁₂ malabsorption.

Lactic acidosis can sometimes occur with metformin therapy. It is more likely to occur in conditions of tissue hypoxia when there is increased production of lactic acid and in renal failure when there is decreased clearance of metformin. Almost all reported cases have involved patients with associated risk factors that should have contraindicated its use (kidney, liver, or cardiorespiratory insufficiency; alcoholism). Acute kidney failure can occur rarely in certain patients receiving radiocontrast agents. Metformin therapy should therefore be temporarily halted on the day of radiocontrast administration and restarted a day or two later after confirmation that renal function has not deteriorated. Renal function should be checked at least annually in patients on metformin therapy, and lower doses should be used in the elderly who may have limited renal reserve and in those with eGFR between 30 and 45 mL/min per 1.73 m².

THIAZOLIDINEDIONES

Thiazolidinediones act to decrease insulin resistance. They are ligands of **peroxisome proliferator-activated receptor gamma (PPAR-γ)**, part of the steroid and thyroid superfamily of nuclear receptors. These PPAR receptors are found in muscle, fat, and liver. PPAR-γ receptors modulate the expression of the genes involved in lipid and glucose metabolism, insulin signal transduction, and adipocyte and other tissue differentiation. Observed effects of the thiazolidinediones include increased glucose transporter expression (GLUT 1 and GLUT 4), decreased free fatty acid levels, decreased hepatic glucose output, increased adiponectin and decreased release of resistin from adipocytes, and increased differentiation of preadipocytes to adipocytes. Thiazolidinediones have also been shown to decrease levels of plasminogen activator inhibitor type 1, matrix metalloproteinase 9, C-reactive protein, and interleukin 6. Two thiazolidinediones are currently available: pioglitazone and rosiglitazone. Their distinct side chains create differences in therapeutic action, metabolism, metabolite profile, and adverse effects. An earlier compound, troglitazone, was withdrawn from the market because of hepatic toxicity thought to be related to its side chain.

Pioglitazone has some PPAR-α as well as PPAR-γ activity. It is absorbed within 2 hours of ingestion; although food may delay uptake, total bioavailability is not affected. Absorption is decreased with concomitant use of bile acid sequestrants. Pioglitazone is metabolized by CYP2C8 and CYP3A4 to active metabolites. The bioavailability of numerous other drugs also degraded by these enzymes may be affected by pioglitazone therapy, including estrogen-containing oral contraceptives; additional methods of contraception are advised. Pioglitazone may be taken once daily; the usual starting dosage is 15–30 mg/d, and the maximum is 45 mg/d. Pioglitazone is approved as a monotherapy and in combination with metformin, sulfonylureas, and insulin for the treatment of type 2 diabetes.

Rosiglitazone is rapidly absorbed and highly protein bound. It is metabolized in the liver to minimally active metabolites, predominantly by CYP2C8 and to a lesser extent by CYP2C9. It is administered once or twice daily; 2–8 mg is the usual total dosage. Rosiglitazone is approved for use in type 2 diabetes as monotherapy, in double combination therapy with a biguanide or sulfonylurea, or in quadruple combination with a biguanide, sulfonylurea, and insulin.

The combination of a thiazolidinedione and metformin has the advantage of not causing hypoglycemia.

These drugs also have some additional effects apart from glucose lowering. Pioglitazone lowers triglycerides and increases high-density lipoprotein (HDL) cholesterol without affecting total cholesterol and low-density lipoprotein (LDL) cholesterol. Rosiglitazone increases total cholesterol, HDL cholesterol, and LDL cholesterol but does not have significant effect on triglycerides. These drugs have been shown to improve the biochemical and histologic features of nonalcoholic fatty liver disease. They seem to have a positive effect on endothelial function: pioglitazone reduces neointimal proliferation after coronary stent placement, and rosiglitazone has been shown to reduce microalbuminuria.

Safety concerns and troublesome side effects have significantly reduced the use of this class of drugs. A meta-analysis of 42 randomized clinical trials with rosiglitazone suggested that this drug increased the risk of angina pectoris or myocardial infarction. As a result, its use was suspended in Europe and severely restricted in the United States. A subsequent large prospective clinical trial (the RECORD study) failed to confirm the meta-analysis finding and so the United States restrictions have been lifted. The drug remains unavailable in Europe.

Fluid retention occurs in about 3–4 % patients on thiazolidinedione monotherapy and occurs more frequently (10–15%) in patients on concomitant insulin therapy. Heart failure can occur, and the drugs are contraindicated in patients with New York Heart Association class III and IV cardiac status (see [Chapter 13](#)). Macular edema is a rare adverse effect that improves when the drug is discontinued. Loss of bone mineral density and increased atypical extremity bone fractures in women are described for both compounds; this is postulated to be due to decreased osteoblast formation. Other adverse effects include anemia, which might be due to a dilutional effect of increased plasma volume rather than a reduction in red cell mass. Weight gain occurs, especially when used in combination with a sulfonylurea or insulin. Some of the weight gain is fluid retention but there is also an increase in total fat mass. In preclinical trials, bladder tumors were observed in male rats on pioglitazone. Initial clinical reports indicated that this might also be true in humans. A 10-year observational cohort study of patients taking pioglitazone, however, failed to find an association with bladder cancer. A large multi-population pooled analysis (1.01 million persons over 5.9 million person-years) also failed to find an association between cumulative exposure of pioglitazone or rosiglitazone and incidence of bladder cancer. Another population based study generating 689,616 person-years of follow-up did find that pioglitazone but not rosiglitazone was associated with an increased risk of bladder cancer.

Troglitazone, the first medication in this class, was withdrawn because of cases of fatal liver failure. Although rosiglitazone and pioglitazone have not been reported to cause liver injury, the drugs are not recommended for use in patients with active liver disease or pretreatment elevation of alanine aminotransferase (ALT) 2.5 times greater than normal. Liver function tests should be performed prior to initiation of treatment and periodically thereafter.

DRUGS THAT AFFECT ABSORPTION OF GLUCOSE

The **α -glucosidase inhibitors** competitively inhibit the intestinal α -glucosidase enzymes and reduce post-meal glucose excursions by delaying the digestion and absorption of starch and disaccharides. **Acarbose** and **miglitol** are available in the United States. **Voglibose** is available in Japan, Korea, and India. Acarbose and miglitol are potent inhibitors of glucoamylase, α -amylase, and sucrase but have less effect on isomaltase and hardly any on trehalase and lactase. Acarbose has the molecular mass and structural features of a tetrasaccharide and very little is absorbed. In contrast, miglitol has structural similarity to glucose and is absorbed.

Acarbose treatment is initiated at a dosage of 50 mg twice daily with gradual increase to 100 mg three times a day. It lowers postprandial glucose levels by 30–50%. Miglitol therapy is initiated at a dosage of 25 mg three times a day. The usual maintenance dosage is 50 mg three times a day, but some patients may need 100 mg three times a day. The drug is not metabolized and is cleared by the kidney. It should not be used in renal failure.

Prominent adverse effects of α -glucosidase inhibitors include flatulence, diarrhea, and abdominal pain and result from the appearance of undigested carbohydrate in the colon that is then fermented into short-chain fatty acids, releasing gas. These adverse effects tend to diminish with ongoing use because chronic exposure to carbohydrate induces the expression of α -glucosidase in the jejunum and ileum, increasing distal small intestine glucose absorption and minimizing the passage of carbohydrate into the colon. Although not a problem with monotherapy or combination therapy with a biguanide, hypoglycemia may occur with concurrent sulfonylurea treatment. Hypoglycemia should be treated with glucose (dextrose) and not sucrose, whose breakdown may be blocked. An increase in hepatic aminotransferases has been noted in clinical trials with acarbose, especially with dosages greater than 300 mg/d. The abnormalities resolve on stopping the drug.

These drugs are infrequently prescribed in the United States because of their prominent gastrointestinal adverse effects and relatively modest glucose-lowering benefit.

DRUGS THAT MIMIC INCRETIN EFFECT OR PROLONG INCRETIN ACTION

An oral glucose load provokes a higher insulin response compared with an equivalent dose of glucose given intravenously. This is because the oral glucose causes a release of gut hormones (“incretins”), principally GLP-1 and glucose-dependent insulinotropic peptide (GIP), that amplify the glucose-induced insulin secretion. When GLP-1 is infused in patients with type 2 diabetes, it stimulates insulin release and lowers glucose levels. The GLP-1 effect is glucose dependent in that the insulin release is more pronounced when glucose levels are elevated but less so when glucose levels are normal. For this reason, GLP-1 has a lower risk for hypoglycemia than the sulfonylureas. In addition to its insulin stimulatory effect, GLP-1 has a number of other biologic effects. It suppresses glucagon secretion, delays gastric emptying, and reduces apoptosis of human islets in culture. In animals, GLP-1 inhibits feeding by a central nervous system mechanism. Type 2 diabetes patients on GLP-1 therapy are less hungry. It is unclear whether this is mainly related to the deceleration of gastric emptying or whether there is a central nervous system effect as well.

GLP-1 is rapidly degraded by dipeptidyl peptidase 4 (DPP-4) and by other enzymes such as endopeptidase 24.11 and is also cleared by the kidney. The native peptide therefore cannot be used therapeutically. One approach to this problem has been to develop metabolically stable analogs or derivatives of GLP-1 that are not subject to the same enzymatic degradation or renal clearance. Four such GLP-1 receptor agonists, exenatide, liraglutide, albiglutide, and dulaglutide are available for clinical use. The other approach has been to develop inhibitors of DPP-4 and prolong the action of endogenously released GLP-1 and GIP. Four oral DPP-4 inhibitors, sitagliptin, saxagliptin, linagliptin, and alogliptin, are available in the United States. An additional inhibitor, vildagliptin, is available in Europe.

GLUCAGON-LIKE PEPTIDE-1 (GLP-1) RECEPTOR AGONISTS

Exenatide, a derivative of the exendin-4 peptide in Gila monster venom, has a 53% homology with native GLP-1 and a glycine substitution to reduce degradation by DPP-4. Exenatide is approved as an injectable, adjunctive therapy in persons with type 2 diabetes treated with metformin or metformin plus sulfonylureas who still have suboptimal glycemic control.

Exenatide is dispensed as fixed-dose pens (5 mcg and 10 mcg). It is injected subcutaneously within 60 minutes before breakfast and dinner. It reaches a peak concentration in approximately 2 hours with a duration of action of up to 10 hours. Therapy is initiated at 5 mcg twice daily for the first month and if tolerated can be increased to 10 mcg twice daily. Exenatide LAR is a once-weekly preparation that is dispensed as a powder (2 mg). It is suspended in the provided diluent just prior to injection. When exenatide is added to preexisting sulfonylurea therapy, the oral hypoglycemic dosage may need to be decreased to prevent hypoglycemia. The major adverse effect is nausea (about 44% of users), which is dose dependent and declines with time. Exenatide monotherapy and combination therapy results in HbA_{1c} reductions of 0.2–1.2%. Weight loss in the range of 2–3 kg occurs and contributes to the improvement of glucose control. In comparative trials the long-acting (LAR) preparation lowers the HbA_{1c} level a little more than the twice-daily preparation. Exenatide undergoes glomerular filtration, and the drug is not approved for use in patients with estimated GFR of less than 30 mL/min.

High-titer antibodies against exenatide develop in about 6% of patients, and in half of these patients an attenuation of glycemic response has been seen.

Liraglutide is a soluble fatty acid-acylated GLP-1 analog. The half-life is approximately 12 hours, permitting once-daily dosing. It is approved in patients with type 2 diabetes who achieve inadequate control with diet and exercise and are receiving concurrent treatment with metformin, sulfonylureas, or thiazolidinediones. Treatment is initiated at 0.6 mg and increased after 1 week to 1.2 mg daily. If needed the dosage can be increased to 1.8 mg daily. In clinical trials liraglutide results in a reduction of HbA_{1c} of 0.8–1.5%; weight loss ranges from none to 3.2 kg. The most frequent adverse effects are nausea (28%) and vomiting (10%). Liraglutide at a dose of 3 mg daily has been approved for weight loss.

Albiglutide is a human GLP-1 dimer fused to human albumin. The half-life of albiglutide is about 5 days and a steady state is reached after 4–5 weeks of once weekly administration. The usual dose is 30 mg weekly by subcutaneous injection. The drug is supplied in a self-injection pen containing a powder that is reconstituted just prior to administration. Weight loss is much less common than with exenatide and liraglutide. The most frequent adverse effects were nausea and injection-site erythema.

Dulaglutide consists of two GLP-1 analog molecules covalently linked to an Fc fragment of human IgG4. The GLP-1 molecule has amino acid substitutions that resist DPP-4 action. The half-life of dulaglutide is about 5 days. The usual dose is 0.75 mg weekly by subcutaneous injection. The maximum recommended dose is 1.5 mg weekly. The most frequent adverse reactions were nausea, diarrhea, and vomiting.

All of the GLP-1 receptor agonists may increase the risk of pancreatitis. Patients on these drugs should be counseled to seek immediate medical care if they experience unexplained persistent severe abdominal pain. Cases of renal impairment and acute renal injury have been reported in patients taking exenatide. Some of these patients had preexisting kidney disease or other risk factors for renal injury. A number of them reported having nausea, vomiting, and diarrhea and it is possible that volume depletion contributed to the development of renal injury. Both exenatide and liraglutide stimulate thyroidal C-cell (parafollicular) tumors in rodents. Human thyroidal C cells express very few GLP-1 receptors, and the relevance to human therapy is unclear. The drugs, however, should not be used in persons with a past medical or family history of medullary thyroid cancer or multiple endocrine neoplasia (MEN) syndrome type 2.

DIPEPTIDYL PEPTIDASE 4 (DPP-4) INHIBITORS

Sitagliptin is given orally as 100 mg once daily, has an oral bioavailability of over 85%, achieves peak concentrations within 1–4 hours, and has a half-life of approximately 12 hours. It is primarily excreted in the urine, in part by active tubular secretion of the drug. Hepatic metabolism is limited and mediated largely by the cytochrome CYP3A4 isoform and, to a lesser degree, by CYP2C8. The metabolites have insignificant activity. Dosage should be reduced in patients with impaired renal function (50 mg if estimated GFR is 30–50 mL/min and 25 mg if <30 mL/min). Sitagliptin has been studied as monotherapy and in combination with metformin, sulfonylureas, and thiazolidinediones. Therapy with sitagliptin has resulted in HbA_{1c} reductions of 0.5–1.0%.

Common adverse effects include nasopharyngitis, upper respiratory infections, and headaches. Hypoglycemia can occur when the drug is combined with insulin secretagogues or insulin. There have been postmarketing reports of acute pancreatitis (fatal and nonfatal) and severe allergic and hypersensitivity reactions. Sitagliptin should be immediately discontinued if pancreatitis or allergic and hypersensitivity reactions occur.

Saxagliptin is given orally as 2.5–5 mg daily. The drug reaches maximal concentrations within 2 hours (4 hours for its active metabolite). It is minimally protein bound and undergoes hepatic metabolism by CYP3A4/5. The major metabolite is active, and excretion is by both renal and hepatic pathways. The terminal plasma half-life is 2.5 hours for saxagliptin and 3.1 hours for its active metabolite. Dosage adjustment is recommended for individuals with renal impairment and concurrent use of strong CYP3A4/5 inhibitors such as antiviral, antifungal, and certain antibacterial agents. It is approved as monotherapy and in combination with biguanides, sulfonylureas, and thiazolidinediones. During clinical trials, mono- and combination therapy with saxagliptin resulted in an HbA_{1c} reduction of 0.4–0.9%.

Adverse effects include an increased rate of infections (upper respiratory tract and urinary tract), headaches, and hypersensitivity reactions (urticaria, facial edema). The dosage of a concurrently administered insulin secretagogue or insulin may need to be lowered to prevent hypoglycemia. Saxagliptin may increase the risk of heart failure. In a postmarketing study of 16,492 type 2 diabetes patients, there were 289 cases of heart failure in the saxagliptin group (3.5%) and 228 cases in the placebo group (2.8%)—a hazard ratio of 1.27. Patients at the highest risk for failure were those who had a history of heart failure or had elevated levels of N-terminal of the prohormone brain natriuretic peptide (NT-pBNP) or had renal impairment.

Linagliptin lowers HbA_{1c} by 0.4–0.6% when added to metformin, sulfonylurea, or pioglitazone. The dosage is 5 mg daily orally and, since it is primarily excreted via the bile, no dosage adjustment is needed in renal failure.

Adverse reactions include nasopharyngitis and hypersensitivity reactions (urticaria, angioedema, localized skin exfoliation, bronchial hyperreactivity). The risk of pancreatitis may be increased.

Alogliptin lowers HbA_{1c} by about 0.5–0.6% when added to metformin, sulfonylurea, or pioglitazone. The usual dose is 25 mg orally daily. The 12.5-mg dose is used in patients with calculated creatinine clearance of 30 to 60 mL/min; the dose is 6.25 mg for clearance <30 mL/min. In clinical trials, pancreatitis occurred in 11 of 5902 patients on alogliptin (0.2%) and in 5 of 5183 patients receiving all comparators (<0.1%). There have been reports of hypersensitivity reactions (anaphylaxis, angioedema, Stevens-Johnson syndrome). Cases of hepatic failure have been reported, but it is unclear if alogliptin was the cause. The medication, however, should be discontinued in the event of liver failure.

Vildagliptin (not available in the United States) lowers HbA_{1c} levels by 0.5–1% when added to the therapeutic regimen of patients with type 2 diabetes. The dosage is 50 mg orally once or twice daily. Adverse reactions include upper respiratory infections, nasopharyngitis, dizziness, and headache. Rarely, it can cause hepatitis, and liver function tests should be performed quarterly in the first year of use and periodically thereafter.

In animal studies, high doses of DPP-4 inhibitors and GLP-1 agonists cause expansion of pancreatic ductal glands and generation of premalignant

pancreatic intraepithelial (PanIN) lesions that have the potential to progress to pancreatic adenocarcinoma. The relevance to human therapy is unclear and currently there is no evidence that these drugs cause pancreatic cancer in humans.

SODIUM-GLUCOSE CO-TRANSPORTER 2 (SGLT2) INHIBITORS

Glucose is freely filtered by the renal glomeruli and is reabsorbed in the proximal tubules by the action of sodium-glucose transporters (SGLTs). Sodium-glucose transporter 2 (SGLT2) accounts for 90% of glucose reabsorption, and its inhibition causes glycosuria and lowers glucose levels in patients with type 2 diabetes. SGLT2 inhibitors lower glucose levels by changing the renal threshold and not by insulin action. The SGLT2 inhibitors canagliflozin, dapagliflozin, and empagliflozin, all oral medications, are approved for clinical use.

Canagliflozin reduces the threshold for glycosuria from a plasma glucose threshold of approximately 180 mg/dL to 70–90 mg/dL. It has been shown to reduce HbA_{1c} by 0.6–1% when used alone or in combination with other oral agents or insulin. It also results in modest weight loss of 2–5 kg. The usual dosage is 100 mg daily. Increasing the dosage to 300 mg daily in patients with normal renal function can lower the HbA_{1c} by an additional 0.5%.

Dapagliflozin reduces HbA_{1c} by 0.5–0.8% when used alone or in combination with other oral agents or insulin. It also results in modest weight loss of about 2–4 kg. The usual dosage is 10 mg daily, but 5 mg daily is recommended initially in patients with hepatic failure.

Empagliflozin reduces HbA_{1c} by 0.5–0.7% when used alone or in combination with other oral agents or insulin. It also results in modest weight loss of 2–3 kg. The usual dosage is 10 mg daily, but 25 mg/d may be used. In a postmarketing multinational study of 7020 type 2 patients with known cardiovascular disease, the addition of empagliflozin was associated with a lower primary composite outcome of death from cardiovascular causes, nonfatal myocardial infarction, or nonfatal stroke (hazard ratio, 0.86; $p = 0.04$). The mechanisms regarding this benefit remain unclear. Weight loss, lower blood pressure, and diuresis may have played a role since there were fewer deaths from heart failure in the treated group whereas the rates of myocardial infarction were unaltered.

As might be expected, the efficacy of the SGLT2 inhibitors is reduced in chronic kidney disease. Canagliflozin and empagliflozin are contraindicated in patients with estimated GFR less than 45 mL/min per 1.73 m². Dapagliflozin is not recommended for use in patients with estimated GFR less than 60 mL/min per 1.73 m². The main adverse effects are increased incidence of genital infections and urinary tract infections affecting about 8–9% of patients. The osmotic diuresis can also cause intravascular volume contraction and hypotension. Canagliflozin and empagliflozin caused a modest increase in LDL cholesterol levels (4–8%). In clinical trials patients taking dapagliflozin had higher rates of breast cancer (nine cases versus none in comparator arms) and bladder cancer (nine cases versus one in placebo arm). These cancer rates exceeded the expected rates in an age-matched reference diabetes population. Canagliflozin has been reported to cause a decrease in bone mineral density at the lumbar spine and the hip. In a pooled analysis of 8 clinical trial (mean duration 68 weeks), an increase in fractures by about 30% was observed in patients on canagliflozin. It is likely that the effect on the bones is a class effect and not restricted to canagliflozin. A modest increase in upper limb fractures was observed with canagliflozin therapy. It is not known if this is due to an effect on bone strength or related to falls due to hypotension. Interim analysis of the Canagliflozin Cardiovascular Assessment Study clinical trial reported an approximately doubled risk of leg and foot amputations in the trial group assigned to Canagliflozin; in 2017 the FDA issued a drug safety communication regarding the association. Cases of diabetic ketoacidosis have been reported with off-label use of SGLT2 inhibitors in patients with type 1 diabetes. Type 1 patients are taught to give less insulin if their glucose levels are not elevated. Because type 1 patients on an SGLT2 inhibitor may have normal glucose levels, they may either withhold or reduce their insulin doses to such a degree as to induce ketoacidosis. Therefore, SGLT2 inhibitors should not be used in patients with type 1 diabetes and in those patients labelled as having type 2 diabetes but who are very insulin deficient and prone to ketosis.

OTHER HYPOGLYCEMIC DRUGS

Pramlintide is an islet amyloid polypeptide (IAPP, amylin) analog. IAPP is a 37-amino-acid peptide present in insulin secretory granules and secreted with insulin. It has approximately 46% homology with the calcitonin gene-related peptide (CGRP; see [Chapter 17](#)) and physiologically acts as a negative feedback on insulin secretion. At pharmacologic doses, IAPP reduces glucagon secretion, slows gastric emptying by a vagally mediated mechanism, and centrally decreases appetite. Pramlintide is an IAPP analog with substitutions of proline at positions 25, 28, and 29. These modifications make pramlintide soluble, non-self-aggregating, and suitable for pharmacologic use. Pramlintide is approved for use in insulin-treated type 1 and type 2 patients who are unable to achieve their target postprandial blood glucose levels. It is rapidly absorbed after subcutaneous administration; levels peak within 20 minutes, and the duration of action is not more than 150 minutes. It is metabolized and excreted by the kidney, but even at low creatinine

clearance there is no significant change in bioavailability. It has not been evaluated in dialysis patients.

Pramlintide is injected immediately before eating; dosages range from 15 to 60 mcg subcutaneously for type 1 patients and from 60 to 120 mcg for type 2 patients. Therapy with this agent should be initiated at the lowest dosage and titrated upward. Because of the risk of hypoglycemia, concurrent rapid- or short-acting mealtime insulin dosages should be decreased by 50% or more. Pramlintide should always be injected by itself using a separate syringe; it cannot be mixed with insulin. The major adverse effects of pramlintide are hypoglycemia and gastrointestinal symptoms, including nausea, vomiting, and anorexia. Since the drug slows gastric emptying, recovery from hypoglycemia can be problematic because of the delay in absorption of fast-acting carbohydrates.

Selected patients with type 1 diabetes who have problems with postprandial hyperglycemia can use pramlintide effectively to control the glucose rise especially in the setting of a high-carbohydrate meal. The drug is not very useful in type 2 patients who can instead use the GLP-1 receptor agonists.

Colesevelam hydrochloride, the bile acid sequestrant and cholesterol-lowering drug, is approved as an antihyperglycemic therapy for persons with type 2 diabetes who are taking other medications or have not achieved adequate control with diet and exercise. The exact mechanism of action is unknown but presumed to involve an interruption of the enterohepatic circulation and a decrease in farnesoid X receptor (FXR) activation. FXR is a nuclear receptor with multiple effects on cholesterol, glucose, and bile acid metabolism. Bile acids are natural ligands of the FXR. Additionally, the drug may impair glucose absorption. In clinical trials, it lowered the HbA_{1c} concentration 0.3–0.5%. Adverse effects include gastrointestinal complaints (constipation, indigestion, flatulence). It can also exacerbate the hypertriglyceridemia that commonly occurs in people with type 2 diabetes.

Bromocriptine, the dopamine agonist, in randomized placebo-controlled studies lowered HbA_{1c} by 0–0.2% compared with baseline and by 0.4–0.5% compared with placebo. The mechanism by which it lowers glucose levels is not known. The main adverse events are nausea, fatigue, dizziness, vomiting, and headache.

Colesevelam and bromocriptine have very modest efficacy in lowering glucose levels, and their use for this purpose is questionable.